

# Module 01: Systems in Embodied

## Learning Objectives

1. Define **Systems** within the context of Embodied.
2. Analyze how **Systems** interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of **Systems**.

## Introduction

This module explores **Systems**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. **Systems** is a critical component of the 8-part Active Inference spine, bridging the gap between [Previous Topic] and [Next Topic].

## Key Concepts

### 1. Systems as a Markov Blanket Boundary

How does **Systems** define the boundary between the agent and the environment?

### 2. Generative Models of Systems

What parameters involved in **Systems** must be optimized to minimize variational free energy?

### 3. Active Inference Dynamics

How does the process of **Systems** drive the perception-action loop?

## Applications

In Embodied, we see **Systems** manifest in: \* **Specific Example 1**: [Add domain-specific example here] \* **Specific Example 2**: [Add domain-specific example here]

## **Conclusion**

Understanding Systems allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.